Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 37: Dots and Lines

Graphs, Degree, and the First Theorem

Chartrand & Zhang, Section 12.1, pp. 432–444

Name: _____ Date: _____

The last new unit

A graph is a set of objects together with a record of which pairs are connected. That is all — and it is enough to model road networks, circuits, social ties, course prerequisites, and the web.
Notice the shape of the definition: a set, plus a set of *pairs*. You have had both since Week 4.

By the end of today you should be able to

1. Give the order, size, and degree sequence of a graph.
2. State and use the First Theorem of Graph Theory.
3. Decide whether a proposed degree sequence is possible.

1. The definition

Graph

A **graph** G consists of a finite nonempty set $V(G)$ of **vertices** and a set $E(G)$ of **edges**, each edge being a 2-element subset of $V(G)$.
The **order** of G is $|V(G)|$; the **size** is $|E(G)|$.

Since an edge is a 2-element subset, no vertex joins itself and no pair is joined twice. The edge $\{u, v\}$ is written uv, and then u and v are **adjacent**.

Work with this graph. $V = \{v_1, \ldots, v_5\}$ and $E = \{v_1v_2,\ v_1v_3,\ v_2v_4,\ v_3v_4,\ v_4v_5\}$. *Draw it in the margin.*

Order: 5 Size: 5

How many graphs are there?

A graph on n labelled vertices is a *subset* of the possible edges, and there are $\binom{n}{2}$ possible edges. So the number of graphs on n vertices is $2^{\binom{n}{2}}$.
For $n = 4$ that is $2^6 = 64$. Power sets from Week 4, combinations from Week 13.

2. Degree

Degree

The **degree** of a vertex v, written $\deg v$, is the number of edges incident with v — equivalently, the number of vertices adjacent to v.
A vertex of degree 0 is **isolated**; a vertex of degree 1 is an **end-vertex** or **leaf**.

For the graph above:

v	v_1	v_2	v_3	v_4	v_5
$\deg v$	2	2	2	3	1

Sum of degrees: 10. Size: 5. Notice anything?

3. The First Theorem of Graph Theory

Theorem 12.10

If G is a graph of order n and size m with $V(G) = \{v_1, \ldots, v_n\}$, then

$$\sum_{i=1}^{n} \deg v_i = 2m.$$

Proof. Summing the degrees counts each edge twice — once at each of its two endpoints. ■

A one-line proof, and it is a counting argument: count the pairs (vertex, incident edge) two ways. That is Lecture 36's technique.

Corollary 12.12

Every graph contains an **even number** of odd vertices.

Why. The total degree $2m$ is even. The even-degree vertices contribute an even amount. So the odd-degree vertices must contribute an even amount too — and a sum of odd numbers is even only when there are an even number of them. ■

4. Ruling things out

Example 12.11. A graph of size 29 has three vertices of each of the degrees 3, 5, and 6, and all remaining vertices have degree 4. What is its order?

Let x be the number of degree-4 vertices. By Theorem 12.10,

$$3(3) + 3(5) + 3(6) + 4x = 2(29) = 58,$$

so $42 + 4x = 58$ and $x = 4$. The order is $9 + 4 = 13$.

Now decide: does such a graph exist?

Proposed degrees	Possible?	Why
$1, 3, 3, 3$	Yes	Sum 10, even; four odd vertices, even count.
$1, 2, 2, 3, 3, 4$	No	Sum 15 is odd — cannot equal $2m$.
A 5-regular graph of order 9	No	Sum 45 is odd.
A 3-regular graph of order 10	Yes	Sum 30, so $m = 15$.

A one-second test

Add the degrees. If the sum is odd, no such graph exists. Equivalently, count the odd degrees — an odd number of them is impossible.
For an r-regular graph of order n the size is $nr/2$, so nr must be even. That rules out every regular graph with both n and r odd.

Exit ticket

1. A graph has size 12. What is the sum of its degrees?

2. Can a graph have degrees $2, 3, 3, 4, 4$? Explain.

Before Wednesday

Read Section 12.1, pp. 436–444 — adjacency matrices, complete and bipartite graphs, isomorphism.

Self-check (not collected). Section 12.1, Exercises 1, 3, 5, and 7 — odd-numbered, answers in the back.

Coming up

Homework 13 is due Friday, November 20, 11:59 p.m.
No quiz next week — Thanksgiving. Quiz 11 was the last one.

Questions I still have